

A Multi-Modal MCP Precision Oncology Platform for High-Grade Serous Ovarian Cancer: Proof-of-Concept

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19 MCP Servers	6 Beyond-SOC Hypotheses (3 HGSOC + 3 ER+ BC)	5 Pipeline Stages
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Abstract

Background. High-grade serous ovarian cancer (HGSOC) has virtually no approved immunotherapy options, and integrated multi-modal computational platforms that systematically identify actionable targets remain lacking.

Methods. We describe proof-of-concept validation of a 19-server Model Context Protocol (MCP) platform executing five sequential stages--genomic profiling, spatial deconvolution, target scoring, causal inference, and report generation--on three synthetic patients: HGSOC (PAT001, Stage IV), ER+ breast cancer (PAT002), and preventive cardiovascular health in a post-menopausal woman (PAT003).

Results. All 14 testable servers passed live validation. For PAT001: POLE p.V411L hypermutation corrected TMB from 4.2 to 47.3 mut/Mb (above FDA pembrolizumab threshold), yielding 568 predicted neoantigens (antigen presentation score 0.94/1.0); spatial deconvolution identified 18.2% CAF fraction and 5.5x CD8 exclusion gradient; Open Targets scored AKT2 (0.373) and CCNE1 (0.279) as top novel targets; 8 actively recruiting trials matched the molecular profile. PAT002 confirmed three FDA-approved treatment paths with zero errors and, via Stage 3--4 deep analysis, three investigational hypotheses not reachable by standard ER+ breast cancer workup: inavolisib preferred over alpelisib for PIK3CA H1047R (2024 FDA approval, superior tolerability); MYC-driven triple therapy (ribociclib + inavolisib + fulvestrant targeting MYC Z=26.9 upstream activation); and cold-TME conversion via YSAPLSSSL neoepitope vaccine (GATA3 frameshift, HLA-B*07:02, IC50 480 nM -- weak binder; threshold <50 nM for strong binding -- vaccine-context priming of exhausted CD8 T cells justifies pursuit despite reduced affinity) + CAF depletion + anti-PD-1, supported by convergent spatial, quantum, and perturbation evidence. Three PAT001 investigational hypotheses--personalized neoantigen vaccine (TP53 R175H -> RMPEAAPPV, IC50 7.8 nM), NNMT/CAF inhibition, and convergent checkpoint blockade--were not reachable by standard workup. For PAT003: cardiometabolic risk profiling identified intermediate 10-year CVD risk (Reynolds 14.3%, Framingham 12.0%, ASCVD 10.3%) with three high-priority evidence gaps (Lp(a), APOE genotype, coronary artery calcium score) not captured by standard lipid panels or population genetic screening.

Conclusions. A 19-server MCP architecture executes comprehensive, multi-modal precision medicine analysis across oncology and preventive health in a single automated pipeline, surfacing clinically actionable hypotheses beyond standard-of-care workup. The addition of a preventive cardiovascular server (PAT003) demonstrates that the architecture generalizes beyond cancer to population health use

cases. Prospective validation on real patient cohorts is the required next step.

***Keywords:** high-grade serous ovarian cancer; model context protocol; precision oncology; neoantigen; spatial transcriptomics; immune exclusion; cancer-associated fibroblasts; NNMT; cross-cancer validation*

1. Introduction

High-grade serous ovarian cancer accounts for approximately 70% of all ovarian cancer deaths and is diagnosed at late stage (IIIC/IV) in the majority of patients. Despite initial response rates of 70--80% to platinum-based chemotherapy, nearly all patients with advanced disease relapse; platinum-resistant disease carries median overall survival under 12 months.

Immunotherapy has transformed outcomes in other solid tumors, yet HGSOC remains highly resistant to immune checkpoint blockade, with objective response rates below 10% in unselected patients. The immune microenvironment is characterized by spatial CD8+ T cell exclusion--cells are present in ascites and stroma but fail to infiltrate the tumor core--a pattern not captured by standard TMB testing. Standard workup (germline BRCA1/2, HRD score, CT staging) guides PARP inhibitors, bevacizumab, and platinum chemotherapy but generates no immunotherapy hypotheses except in the rare MSI-H subgroup (<1%).

The Model Context Protocol (MCP), introduced by Anthropic in 2024, provides a structured mechanism for orchestrating heterogeneous tools through a common interface, enabling agentic pipelines that chain genomic databases, spatial transcriptomics engines, neoantigen predictors, and clinical trial registries in a single automated workflow. Here we demonstrate that a 19-server MCP architecture executing 104 tool calls across 14 servers can surface three clinically actionable treatment hypotheses for HGSOC that standard workup cannot reach, generalize across cancer types, and extend to preventive cardiovascular health without disease-specific code changes.

2. Methods

2.1 Platform Architecture

The platform comprises 19 MCP servers, each wrapping a distinct data source or computational tool, orchestrated by Claude Sonnet 4.6 via a structured workflow prompt. For oncology patients (PAT001, PAT002) outputs propagate as structured JSON through five sequential stages; for preventive health (PAT003) the cardiometabolic server executes a parallel risk-stratification workflow. Outputs propagate as structured JSON between stages (Figure 1).

[FIGURE 1 PLACEHOLDER -- Insert pipeline diagram here]

Figure 1. Five-stage MCP precision oncology pipeline. Patient data flows left to right from intake through report generation.

Stage	Description	Servers	Key Outputs
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1\.. Data Acquisition	Patient records, FASTQ QC, TCGA, GEO, somatic variants, CNV, HRD	mockepic, fgbio, mocktcga, geodownload, genomic-results	Variant calls, HRD score, expression matrix
2\.. Spatial Deconvolution	Cell-type deconvolution (NNLS), spatial autocorrelation	spatialtools, cibersortx	Cell-type fractions, Moran's I
3\.. Target Profiling	Open Targets scoring, drug linkages, upstream regulators	opentargets, multiomics	Ranked targets, drug candidates
4\.. Causal Inference	GEARS GNN perturbation; quantum fidelity evasion; neoantigen/MHC-I prediction	perturbation, quantum, cell-classify, neoantigen, openimagedata	Perturbation predictions, evasion scores, neoantigens, cross-validated hypotheses
5\.. Report Synthesis	Clinical report generation, trial matching	patient-report, clinicaltrials.gov	PDF report, trial recommendations

2.2 Server Descriptions

Server	Function	Mode
mcp-mockepic	Patient clinical records (FHIR mock)	Synthetic
mcp-fgbio	FASTQ QC and genomic utilities	Live/Mock
mcp-geodownload	NCBI GEO dataset search and download	Live
mcp-genomic-results	VCF/CNS parsing, HRD scoring	File-based
mcp-spatialtools	Spatial QC, deconvolution, visualization	Live
mcp-cibersortx	Immune deconvolution (NNLS; CIBERSORTx REST when token available)	NNLS/REST
mcp-opentargets	Open Targets evidence scores and drugs	Live GraphQL
mcp-neoantigen	Neoantigen burden, IEDB MHC-I binding, HLA typing	Live (IEDB)
mcp-perturbation	GEARS GNN perturbation prediction+	GNN
mcp-quantum	Quantum circuit cell-type embeddings	Live
mcp-clinicaltrials	ClinicalTrials.gov trial matching	Live
mcp-patient-report	PDF patient report generation	Live

mcp-cardiometabolic ++	CVD risk scoring (Reynolds/Framingham/PCE), biomarker panel, statin decision, lifestyle evidence	Live
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+ mcp-perturbation: GEARS GNN trained on synthetic HGSOC-modeled Perturb-seq data (cell-gears 0.1.2; Pearson $r = 0.976$ on unseen singles). GEARS.__init__() hidden_size bug resolved March 28, 2026; all remaining bugs resolved April 17, 2026 (FastMCP 2.x upgrade + anndata nullable strings + GEARS API compatibility). Full live end-to-end pipeline validated April 17, 2026 (no dry_run). Full 19-server specifications in Supplementary Table S1.

+ mcp-perturbation: GEARS GNN trained on synthetic HGSOC-modeled Perturb-seq data (cell-gears 0.1.2; Pearson $r = 0.976$ on unseen singles). GEARS.__init__() hidden_size bug resolved March 28, 2026; all remaining bugs resolved April 17, 2026. Full live end-to-end pipeline validated April 17, 2026 (no dry_run). ++ mcp-cardiometabolic: Reynolds Risk Score (Ridker et al. JAMA 2007; women-specific), Framingham Risk Score (Wilson et al. Circulation 1998), and ACC/AHA PCE (Goff et al. JACC 2014); statin decision logic per 2018 ACC/AHA guidelines. Live PAT003 end-to-end validation April 23, 2026 (no dry_run).

2.3 Proof-of-Concept Patients

PAT001 (HGSOC, Stage IV): 58-year-old female; synthetic case modeled on published TCGA-OV frequencies. Key features:

Feature	PAT001 Value	TCGA-OV Frequency
TP53 mutation	R175H, AF 73%	>96% of HGSOC
BRCA1 germline	Pathogenic (c.5266dupC)	~15% germline BRCA1/2
CCNE1 amplification	$\log_2 = 1.58$	~15--20%; platinum-resistance marker
AKT2 amplification	$\log_2 = 1.15$	~12%; PI3K/AKT pathway
RB1 deletion	$\log_2 = -1.85$	~8%; CDK4/6 inhibitor resistance
POLE p.V411L	Somatic, exonuclease domain	Rare; hypermutator
HRD score	54 (≥ 42 threshold)	~50% of TCGA-OV
TMB (POLE-corrected)	47.3 mut/Mb	Median HGSOC ~2--4 mut/Mb

Spatial fixture: 900 Visium spots across 7 tissue regions; CAF fraction 18.2% (164/900 spots); 9.1% CD8+ T cells (82/900 spots).

PAT002 (ER+/PR+/HER2- IDC, Stage IIA): BRCA2 germline pathogenic variant; PIK3CA H1047R; HRD score 35; ESR1/PGR strongly spatially clustered (Moran's $I = 0.42$ --0.45). Cross-cancer-type generalizability of the pipeline architecture validated via full SYNTHETIC_DATA mode end-to-end run (May 2026).

PAT003 (Preventive Cardiovascular Health): 67-year-old post-menopausal female; synthetic case representative of an underserved demographic in CVD clinical research. Key cardiometabolic values: LDL 118 mg/dL, HDL 58 mg/dL, total cholesterol 195 mg/dL, triglycerides 142 mg/dL, hsCRP 1.8 mg/L, BP 138/84 mmHg (Stage 1 hypertension), HbA1c 5.6%. Helix Tier 1 population genetic screen (negative): monogenic familial hypercholesterolaemia ruled out across LDLR/APOB/LDLRAP1/PCSK9; primary CVD risk mechanism reclassified to polygenic and environmental. Included to demonstrate that

the MCP architecture generalises from oncology to preventive health. Three evidence gaps not captured by standard lipid panels or population screens constitute the platform's primary actionable output: serum Lp(a), APOE genotyping, and coronary artery calcium score.

2.4 Statistical Methods

Spatial autocorrelation: Moran's I with k=6 nearest-neighbor weights, 999 permutations. Cell-type deconvolution: NNLS against an 8-category HGSOC marker signature (Olbrecht et al. 2021, Xu et al. 2022). Multi-omics: Stouffer's Z-score, Benjamini-Hochberg FDR $q < 0.05$. MHC-I binding: NetMHCpan 4.1 via IEDB REST API; strong binder threshold IC50 < 50 nM. HRD: simplified genomic scar (LOH+TAI+LST); proof-of-concept only, not Myriad-validated.

3. Results

3.1 Platform Validation

All 14 servers with testable live tools passed without error in re-validation on April 23, 2026 (original validation: March 9, 2026; PAT001/PAT002 re-validation: April 17, 2026; PAT003 first validation: April 23, 2026); 5 servers operate in synthetic/mock mode by design (see Supplementary Table S1). Total: 104 tools across 19 servers, 0 errors in combined validation. Full live perturbation pipeline (load->setup->train->predict, no dry_run) confirmed passing April 17, 2026 following FastMCP 2.x upgrade and regression resolution. PAT003 cardiometabolic pipeline (5 tools across mcp-cardiometabolic + 7 CVD gene-disease associations via mcp-opentargets, no dry_run) confirmed passing April 23, 2026.

3.2 Stage 1: Genomic Profile

PAT001's somatic variant profile (BRCA1 germline, TP53 R175H, CCNE1 amp log2=1.58, AKT2 amp log2=1.15, RB1 del log2=-1.85) is consistent with the TCGA-OV platinum-resistant stratum. The POLE p.V411L somatic mutation--a proofreading-domain variant--was identified at this stage and carried forward to Stage 4 for TMB correction. HRD score 54 confirms eligibility for PARP inhibitor maintenance. PAT002's BRCA2 germline pathogenic variant triggered the platinum-sensitive/PARP-eligible pathway and PIK3CA H1047R flagged PI3K inhibitor eligibility.

3.3 Stage 2: Spatial Deconvolution

Region	Dominant Cell Type	CD8A (TPM)	EPCAM (TPM)
tumor_core_luminal	Tumor epithelial (EPCAM+)	89	412
stroma_immune	CD8+ T cell / TAM	493	28
stroma_fibrous	Fibroblast/CAF	34	61
tumor_invasive_front	Mixed	187	298

Fibroblast/CAF content: 18.2% (164/900 spots), placing PAT001 in the CAF-high immunosuppressive

stratum. CD8A shows a 5.5x gradient from stroma_immune (493 TPM) to tumor_core (89 TPM), consistent with immune exclusion. Moran's I for CD8A = 0.019 (diffuse, weakly patterned). PAT002's ESR1/PGR showed strong spatial clustering (Moran's I = 0.42--0.45), correctly interpreted as hormone-receptor-driven proliferation.

3.4 Stage 3: Target Profiling

Open Targets batch query returned AKT2 (log2=+1.15, OT score 0.373; Cancer Gene Census component 0.61) and CCNE1 (log2=+1.58, OT score 0.279) as the highest-scoring novel targets not addressed by standard HGSOC workup. Upstream regulator analysis identified STAT3 and MYC as the top transcription factor drivers (activation Z-score > 2.0 after Stouffer meta-analysis across spatial and expression layers).

Co-occurring CCNE1 amplification, AKT2 amplification, and RB1 deletion define a convergent G1/S dysregulation profile associated with combined CDK2+AKT resistance. CCNE1 amplification is the primary CDK2 activator driving S-phase entry and is not assessed in standard HGSOC clinical workup. The co-occurrence of CCNE1 and AKT2 amplification alongside RB1 deletion creates two independent treatment-resistance escape routes, both of which have FDA-IND active trials currently enrolling (NCT06586957: NKT3964 oral CDK2 degrader; NCT05039801: IACS-6274 + capivasertib). The Open Targets CCNE1 score of 0.279 against ovarian carcinoma (EFO_0001071) is non-zero and biologically consistent, though limited by API node resolution tier; AKT2 score of 0.373 is the highest returned in the batch query.

3.5 Stage 4: Causal Inference

POLE correction. POLE p.V411L produces ultrahigh TMB through polymerase proofreading failure: corrected TMB = 47.3 mut/Mb (vs. 4.2 mut/Mb nominal), above the FDA 10 mut/Mb pembrolizumab threshold. Estimated neoantigens: 568 (108 strong MHC-I binders; IC50 < 50 nM); antigen presentation pathway score 0.94/1.0.

Neoantigen. TP53 R175H -> peptide RMPEAAPPV -> HLA-A02:01 binding IC50 = 7.8 nM (*strong binder*). HLA-A02:01 frequency ~45% in Caucasian HGSOC patients.

Perturbation (GEARS GNN). CCNE1 knockdown: CDK2 (-0.42), MYC (-0.25), CDKN2A (+0.31), RB1 (+0.20). NNMT knockdown: STAT3 (-0.24), COL3A1 (-0.21), PRF1 (+0.27), FDX1 (+0.20), indicating CAF barrier dismantlement with immune recovery.

Quantum fidelity. 234/300 spots (78%) flagged as immune evasion states (threshold 0.30), consistent with the spatial CD8 exclusion pattern.

3.6 Cross-Cancer Validation: PAT002

The identical pipeline executed on PAT002 with zero errors, identifying three FDA-approved treatment paths: PARP inhibitor (olaparib 300 mg BID; BRCA2 germline, HRD 35--below myChoice threshold but PARP-eligible via germline route); endocrine + CDK4/6 inhibitor (letrozole + palbociclib; ESR1/PGR+); PI3K inhibitor (alpelisib + fulvestrant; PIK3CA H1047R). The HRD-negative/BRCA-positive nuance was handled without disease-specific code changes, confirming architecture-level cancer-type agnosticism.

PAT002-BC-2026 completed a full SYNTHETIC_DATA mode end-to-end run across five active stages. Genomic analysis parsed 5 somatic variants (PIK3CA H1047R VAF 0.42, GATA3 frameshift VAF 0.31, CDH1 splice site VAF 0.28, MAP3K1 VAF 0.35, TP53 subclonal VAF 0.15) and a breast-cancer-specific

CNV profile (chr17q loss, chr8q/CCND1 gain, HER2 neutral). HRD score of 35 confirms PARP inhibitor eligibility via germline BRCA2. Spatial analysis of 900 Visium spots identified seven cell types with ESR1/PGR enrichment in tumor core, a tertiary lymphoid structure B cell cluster, and spatial HER2 negativity confirmed. Multi-omics Stouffer integration ranked CDK4/6 inhibition, PI3K pathway inhibition, and PARP inhibitor sensitivity as top therapeutic targets. A draft precision oncology report was generated with clinician review gate.

Quantum cell-type fidelity analysis (500 cells, 8 cell types) resolved the immune evasion mechanism as CAF-mediated stromal exclusion rather than MHC-I downregulation or IDO pathway activation. Three signals supported this: uniform exhaustion boost across all 500 cells (mean 0.314); near-identical immune fidelity across all four immune subtypes (CD8 0.628, CD4 0.627, macrophage 0.637, B cell 0.625), indicating global suppression not lineage-specific editing; and the sole high-evasion cell (score 0.811) being a CD8_T_cell in peak exhaustion state -- consistent with chronic failed penetration of a CAF-rich stroma. CAFs represent 20% of the TME (100/500 cells), second only to luminal tumour cells (36%).

Quantum analysis substantially strengthened this hypothesis by confirming that CD8 T cells are present (70/500, 14%) and exhausted -- not absent -- making them a viable substrate for vaccine priming. The convergent four-layer evidence (spatial exclusion, quantum exhaustion, neoantigen MHC binding, perturbation CAF signalling) supports a sequential intervention: (1) stromal barrier dissolution via anti-TGF-beta or CAF depletion (e.g. nintedanib); (2) YSAPLSSSL mRNA neoepitope vaccination to expand GATA3-reactive CD8 clones; (3) anti-PD-1 to prevent re-exhaustion of newly primed effectors. Steps are order-dependent -- vaccination without prior stromal disruption cannot overcome the physical exclusion barrier.

3.7 Preventive Health Validation: PAT003

The cardiometabolic server executed a 5-tool live pipeline on PAT003 with zero errors. Risk scores converged on intermediate 10-year CVD risk: Reynolds 14.3% (women-specific; Ridker et al. JAMA 2007), Framingham 12.0% (Wilson et al. Circulation 1998), ASCVD 10.3% (Goff et al. JACC 2014). All three algorithms independently placed PAT003 above the 7.5% ACC/AHA threshold for statin consideration. hsCRP at 1.8 mg/L fell just below the JUPITER trial criterion (≥ 2.0 mg/L; Ridker et al. NEJM 2008) that confers additional rosuvastatin benefit, identifying a clinically meaningful margin. The biomarker panel flagged LDL as near-optimal, HDL as acceptable, and BP as Stage 1 hypertension. Open Targets returned CVD gene-disease associations for all seven queried targets (APOE 0.816, LDLR 0.737, ACE 0.752, PCSK9 0.726, LPA 0.601, CDKN2A 0.513, CDKN2B 0.536).

The platform's most clinically significant output was the identification of three high-priority evidence gaps absent from standard lipid panels and absent from the Helix Tier 1 population genetic screen (negative result; FH ruled out). First, serum Lp(a) was not measured: Lp(a) is genetically determined, does not respond to statins, and is an independent CVD risk factor; a single lifetime measurement is recommended by 2023 ESC/EAS guidelines. Second, APOE genotype was not assessed: APOE is the strongest common genetic determinant of both CVD and Alzheimer's disease risk, yet it is absent from all population screening panels. Third, coronary artery calcium (CAC) score was not available: at intermediate risk (7.5%--20%) CAC is the best-validated reclassification tool and is endorsed by 2018 ACC/AHA guidelines as the preferred risk enhancer for statin decision-making. The negative genetic screen result--while clinically useful in ruling out monogenic FH--shifted the interpretive frame from "find the causal variant" to "quantify polygenic and biomarker risk," making these three gaps the actionable output.

3.8 Clinical Trial Matching

Eight actively recruiting trials matched PAT001's molecular profile:

NCT ID	Phase	Sponsor	Molecular Match
NCT06580314	III	NRG Oncology	BRCA1 germline + HRD 54; olaparib ± bevacizumab
NCT06586957	I	NiKang Therapeutics	CCNE1 log2=1.58; NKT3964 CDK2 degrader
NCT05867251	I/II	Avenzo Therapeutics	CCNE1 amplification; AVZO-021
NCT06188520	I/II	AstraZeneca	HGSOC-specific; AZD8421
NCT05086692	I/II	Medicenna Therapeutics	TMB 47.3 + POLE; MDNA11 + pembrolizumab
NCT05039801	I	MD Anderson	AKT2 amp + HGSOC; IACS-6274 + capivasertib
NCT03675893	II	Dana-Farber	AKT2 amp; RESOLVE letrozole + abemaciclib
NCT05564377	II	NCI	BRCA1 + AKT2; ComboMATCH olaparib + ipatasertib

The CCNE1-specific trials (NCT06586957, NCT05867251) and the TMB-high trial (NCT05086692) were surfaced without prior hypothesis; CCNE1 copy number is not assessed in standard HGSOC clinical workup.

4. Cross-Validated Treatment Paths vs. Standard of Care

Three investigational paths were identified beyond standard-of-care (Table below). These are not de novo biological discoveries--the underlying biology is documented in existing literature--but they were not reachable by standard workup in a single automated pass.

Treatment	Standard Care?	Platform Trigger	Evidence Level
Carboplatin + Paclitaxel	[Y] Yes	Stage IV diagnosis	NCCN Category 1
Bevacizumab maintenance	[Y] Yes	High-risk features	Level I (ICON7)
Olaparib maintenance	[Y] Yes	HRD 54	Level I (SOLO-1)
Pembrolizumab / checkpoint ICI	[Y] Indicated	TMB 47.3; POLE	Level I (TMB-H)

Neoantigen vaccine (RMPEAAPPV)	[N] Novel	No standard assay	Investigational
NNMT inhibition (CAF targeting)	[N] Novel	No fibroblast assay in SOC	Investigational
Checkpoint + spatial exclusion reversal	[N] Novel	Spatial CD8 pattern	Investigational
PAT002 (ER+ BC)			
Inavolisib over alpelisib	[N] Novel	OT drug profiling + upstream regulator (MYC Z=26.9)	Phase 3 (INAVO010); 2024 FDA approval
MYC triple therapy (ribociclib + inavolisib + fulvestrant)	[N] Novel	MYC upstream Z=26.9; CCND1 Stouffer q=1.1e-5; CDK4 OT 0.637	Investigational (SERENA-6 / MONALEESA framework)
YSAPLSSSL vaccine + CAF depletion + anti-PD-1	[N] Novel	Spatial cold TME + quantum CD8 exhaustion + neoantigen MHC binding	Weak MHC binder (IC50 480 nM); Investigational (no standard ER+ BC immunotherapy pathway)

Finding 1 -- Personalized Neoantigen Vaccine. TP53 R175H generates peptide RMPEAAPPV with HLA-A02:01 binding $IC_{50} = 7.8$ nM (strong binder threshold: <50 nM). TP53 R175H occurs in ~8--12% of HGSOC; HLA-A02:01 in ~45% of Caucasian patients. Standard workup records the mutation but no clinical pathway queries IEDB for binding confirmation; the platform executes this automatically.

Finding 2 -- NNMT/CAF Inhibition. Spatial deconvolution identified 18.2% fibroblast content (CAF-high stratum). GEO search surfaced GSE292142 (2025): NNMT inhibition dismantles the CAF barrier in HGSOC. GEARS perturbation confirms NNMT knockdown reduces STAT3/COL3A1 and increases PRF1/FOXP3, indicating immune recovery. No standard clinical assay measures fibroblast content; this linkage required automated multi-modal integration.

Finding 3 -- Convergent Checkpoint Evidence. Both TMB-based evidence (POLE → 47.3 mut/Mb, regulatory-grade) and spatial evidence (5.5x CD8 exclusion gradient) independently converge on checkpoint blockade. The spatial data adds a clinically actionable prediction: exclusion-reversal co-treatment (CXCR4 blockade, FAK inhibition, anti-TGF-beta) may be required for checkpoint monotherapy to be effective--a hypothesis not derivable from TMB alone.

5. Discussion

Stage 4 of the platform produces a qualitatively different class of evidence from Stages 1--3. Stages 1--3 generate correlative evidence--the same type as current clinical molecular profiling. Stage 4 adds a causal inference layer through GEARS GNN perturbation modeling, quantum cell-type fidelity analysis, and neoantigen binding prediction, each mechanistically orthogonal, independently stress-testing whether a candidate target is therapeutically actionable rather than merely associated.

The platform's contribution is integrative, not biological. All three investigational findings are grounded in existing literature; the platform's clinical value is connecting them automatically from a single patient's

data in one pipeline run, without prior hypotheses about NNMT, CCNE1 CDK2 inhibitor trials, or RMPEAAPPV binding--steps that standard SOC workup cannot replicate.

Generalisation across disease types is demonstrated by PAT002 and PAT003. For PAT002 (ER+ IDC), the identical oncology architecture handled a clinical nuance (HRD 35 < myChoice threshold but BRCA2 germline PARP-eligible) that required no disease-specific code. The contrast in spatial biology between the two oncology patients is instructive: ESR1 and PGR are strongly spatially clustered in PAT002 (Moran's I = 0.42--0.45), indicating hormone-receptor-driven proliferation, whereas PAT001's immune markers showed diffuse, weakly patterned expression (Moran's I = 0.003--0.092). The platform correctly interpreted both patterns in their disease-specific context using the same autocorrelation logic. For PAT003 (preventive cardiovascular health), the cardiometabolic server operates as a parallel risk-stratification pipeline rather than a sequential oncology workflow; its integration required only a new MCP server and JSON schema extension with no changes to the orchestration layer. This extension is clinically motivated: women are significantly underrepresented in the landmark CVD trials that established Framingham risk coefficients (predominantly White male cohort), and the Reynolds Risk Score was specifically validated in women incorporating hsCRP and family history as independent risk enhancers. The MCP server layer functions as the disease-adaptation interface; the orchestration logic is both cancer-type and disease-domain agnostic.

Critically, PAT002 demonstrated not only architectural generalizability but also investigational hypothesis generalizability: the identical Stage 3--4 analytical layers (target profiling, perturbation, quantum, neoantigen) generated three beyond-standard-workup hypotheses for ER+ breast cancer by the same mechanism they generated three for HGSOC -- without disease-specific tuning. This is a stronger claim than pipeline portability: it suggests the platform's capacity to surface investigational hypotheses is a property of the multi-modal integration architecture, not of the specific cancer biology encoded in PAT001.

The spatial immune exclusion finding also highlights a fundamental limitation of current FDA biomarker frameworks: TMB measures mutational diversity as a proxy for neoantigen burden but does not assess whether T cells are excluded from the tumor core or what mechanism drives exclusion. Spatial transcriptomics at the resolution implemented here distinguishes all three scenarios (immune desert, immune exclusion, inflamed). The 5.5x CD8 gradient in PAT001 is consistent with the exclusion pattern characterized by Vázquez-García et al. [1] and independently confirmed by quantum cell-type fidelity analysis (78% spots flagged as evasion states). While Visium is not yet standard of care, its cost has fallen to approximately \$1,500--\$3,000 per sample in research settings, making it a near-term candidate for prospective integration.

Limitation	Impact	Mitigation
Synthetic patient data	Results may reflect parameter choices, not real biology	Features drawn from published TCGA-OV/BRCA frequencies
GEARS trained on synthetic data	Perturbation predictions not validated against real HGSOC CRISPR screens	Pearson $r = 0.976$ on held-out singles; labeled as synthetic-data-derived
No benchmarking vs. standalone tools	Cannot claim performance superiority	Component comparisons (NNLS vs. CARD/RCTD; pVACseq vs. mcp-neoantigen) are planned

HRD calculation not Myriad-validated	Clinical eligibility claims are indicative only	Simplified POC calculation; requires myChoice validation
Real patient data access	HIPAA-compliant federated access required for prospective validation	Terra/AnVIL or institutional EHR platform (e.g., Mayo Clinic Platform) as candidate substrate

Prospective validation on 30--50 real patients with matched clinical outcomes is the single most important next step. Integration with a clinical data infrastructure--such as the Mayo Clinic Platform's de-identified, OMOP-standardized multi-institutional EHR--would address the HIPAA access barrier and provide the real-world evidence needed to support translational deployment.

6. Conclusions

A 19-server MCP orchestration platform can execute comprehensive, multi-modal precision medicine analysis from patient intake through clinical report generation in a single automated pipeline, identifying three clinically actionable treatment hypotheses for HGSOC that standard NCCN-guided workup cannot surface. Cross-disease generalizability was confirmed on ER+ breast cancer (PAT002) and preventive cardiovascular health (PAT003) without platform-level code changes: the cardiometabolic server extended the architecture beyond oncology by implementing women-specific CVD risk algorithms (Reynolds, Framingham, PCE) and surfacing three high-priority evidence gaps--Lp(a), APOE genotype, and coronary artery calcium score--not captured by standard lipid panels or population genetic screening. Prospective validation on real patient cohorts and integration with clinical data infrastructure are the required next steps toward translational deployment.

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Declarations

Ethics Statement. PAT001, PAT002, and PAT003 are fully synthetic. No human subjects data were used; no IRB approval was required.

Data Availability. MCP server source code and synthetic datasets: <https://github.com/lynnlangit/precision-medicine-mcp> (upon publication). Archived API response snapshots from April 17, 2026 re-validation (including live perturbation pipeline run) and April 23, 2026 PAT003 cardiometabolic validation (no dry_run) are included in the repository.

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Author Contributions. L.L. designed the platform, implemented the MCP server infrastructure, conducted all experiments, and wrote the manuscript.

Reproducibility details (computational environment, tool version pins, fixture file hashes, and re-run guidance) are provided in Supplementary Materials (Appendix A).